Islamic University of Lebanon

Faculty of Sciences & Arts – CS – Second Year

Faculty of Economics & Business Administration – MIS – Third year

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Course:	*Relational Database II – Lab*
Academic Year:	*Third Year (5th semester)*
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Reference:

1. .

Database: ShoppingCart

Given the database '**ShoppingCart**' which his relational model is the following:

- **Clients** (Num, Name, Address, Phone)
- **Items** (ID, Name, PurchasePrice, SalePrice, QuantityStock)
- **Invoices** (Num, InvoiceDate, *ClientNum**)
- **DetailInvoice** (*ItemID*, InvoiceNum**, Quantity, InvoicedPrice)
- **FeaturedItem**(*ItemID**, FeatureText, FeaturePrice)

Table	Primary key	Foreign key
Clients	Num	None
Items	ID	None
Invoices	Num	ClientNum refers to the Clients table.
DetailInvoice	ItemID, InvoiceNum	ItemID refers to the Items table. InvoiceNum refers to the Invoices table.
DiscountItem	ItemID	ItemID refers to the Items table.

Suppose we have the following information in the three relations:

Clients Table

Num	Name	Address	Phone
333	Bob Jonson	Vience	03-333333
455	Jack Brown		01-111222
656	Mary Smith	Los Angeles	03-123456
765	Kelly Bay	Oxford	03-123123
234	Al Malak	Los Angeles	07-777000

Invoices Table

Num	InvoiceDate	*ClientNum**
1	1/1/2016	455
2	2/3/2016	333
3	30/6/2017	234
4	8/6/2017	455

Items Table

ID	Name	PurchasePrice	SalePrice	QunatityStock
100	CD-ROM	30	50	30
101	RAM	100	135	10
102	Flash	20	50	13
103	Keyboard	10	17	7

FeatureItem Table

*ItemID**	FeatureText	FeaturePrice
100	Discount	20
101	Offer	100
103	Discount	12

InvoiceDetail Table

*InvoiceNum**	*ItemID**	Quantity	InvoicedPrice
1	100	3	53
1	103	1	17
2	100	1	50
2	102	1	55
3	101	1	135
4	103	3	17

Lab1: Definition of Data

Q1). Use the SQL language to create the *ShoppingCart* database and its tables as presented above. The following constraints must be taken into account:

- **All tables**:
  - The **primary keys** are underlined and the **foreign keys** are in italics followed by the symbol '*'.
- **`Clients`** Table:
  - The Name and phone of client must be known.
  - The phone number of the client have the following format: xx-xxxxxx where x represent a digit.
- **`Items`** Table:
  - The attributes PurchasePrice and SalePrice must always have values > 5.
  - The value of the attribute SalePrice must be greater than or equal to that of PurchasePrice.
  - The value of QuantityStock attribute is greater than or equal to 0.
  - It is impossible to have two items with the same name.
  - The Name, Purchase and Sale Price of Items must be known.
- **`Invoices`** Table:
  - The **`Num`** attribute of table **`Invoices`** is *auto-increment*.
  - The invoice date must be less than or equal to the current date.
  - The default value of the invoice date is the system date.
  - The deletion of a client leads the deletion of all its invoices. Consequently, the deletion of the detail of that invoices.
- **`DetailInvoice`** Table:
  - The value of quantity and InvoicedPrice attributes must be greater than 0.
  - The deletion of a client leads the deletion of all its invoices. Consequently, the deletion of the detail of that invoices.
- **`FeatureItem`** Table:
  - The possible values of the FeatureText attribute are: Discount and Offer.

Lab 2: DDL & DML

Q1). Fill the five tables by the information presented in the previous page.

Q2). **Add** the attribute **Salary (20 characters)** to the **Clients** table.

Q3). **Modify** the **Salary** attribute type of the Clients table to NUMBER.

Q4). The smallest salary of a client is 100. Write the SQL statement that allows to realize that.

Q5). Now, we suppose the salary is increased and the smallest one is 1000. Write the SQL statement(s) that allows to realize that.

Q6). **Remove** the attribute **Salary** from the **Clients** table.

Q7). **Insert** into the **Clients** table the following record: (222, 'Mike Petit', 'Paris').

Q8). **Delete** the client identified by 222.

Q9). **Increase** the **selling price** of **all** items of **10%.** Then, cancel the update.

Q10). Is it possible to insert the following records in the **Order** database? **Justify your answer**.

a. The record (104, 'CD-ROM', 40, 50, 10) into the **Items** table.
b. The record (1, #1/1/2010#, 200) into the **Invoices** table.
c. The record (2, 103, 5) into the **DetailInvoice** table.
d. The record (2, 110, 5) into the **DetailInvoice** table.
e. The record (2, 101, 1) into the **DetailInvoice** table.

Lab 3: SQL Queries

Q1). Apply the following queries on the five tables of Order database (page 1) and **give their results**.

1. `Select * From Clients Where Num = 333.`
2. `Select COUNT(*) as 'Inoices count'`
   `From Invoices`

Q2). Write queries answering the following questions using **SQL**:

1. Show clients in **descending** order of name and **ascending** order of address.
2. Show **all invoices** of the client identified by **455**.
3. Show the **names** of clients whose **first** and **fourth** letters of the name are respectively '**M**' and '**A**'.
4. Display the clients where their names start with M and end with h.
5. List all invoices of client identified by 333 and purchased after 30/1/2014**.**
6. List the invoices purchased in 2014.
7. Display client addresses. Note that an address must be displayed only once.
8. Display all items having a sale price between 30 and 75.
9. List the clients living in Oxford and Los Angeles.
10. Give the total amount of the invoice having 2 as number.
11. Display the clients who do not have an address.
12. Display the clients who have a fixed address.

Lab 4: SQL Queries

Q1). Apply the following queries on the five tables of Order database (page 1) and **give their results**.

1. `Select Name, Count(*) FROM Clients, Invoices WHERE Num = Client Group By Name.`

Q2). Write queries answering the following questions using **SQL**:

1. Give the two items with the highest sale price.
2. Give for **each invoice**, the **number** of purchased items.
3. Give for **each invoice**, the total amount of its items. (Total amount of an item = quantity * sale price).
4. Give the ID, name, sale price, quantity and total for each item in the invoice having 2 as number.
5. **How many clients** buy the item **identified by** 100?
6. Give the items having an **average of selling price greater than or equal to 35**.
7. Display the clients who **have never purchased** an item.
8. Give the clients who **purchased an item purchased by the client identified by 333**.
9. Give the items having the **highest price**.
10. Give the clients who purchased all items.
11. Show the features (ID, name, SalePrice) of all items and their discount prices if possible.

Lab 5: Procedures & Functions

Q1). Write a function that takes an invoice number as a parameter and returns the total of that invoice.

Q2). Write a function that takes an item code in parameter and that returns the final price of this article. The final price is calculated as follows:

- If the quantity in stock <3, then the final price is equal to 40% of the selling price.
- If the quantity in stock is between 3 and 7, then the final price is equal to 60% of the selling price.
- If the quantity in stock is between 8 and 12, then the final price is equal to 90% of the selling price.
- Otherwise, the final price is equal to the selling price.

Q3). Write a procedure that takes a string as a parameter and that displays all items containing the string.

Q4). Write a procedure that takes an invoice number as a parameter and displays the detail of the invoice as follows:

```
Invoice number: 1
Date: 1/1/2016
Code    Name       Quantity    Price    Total
100     CD-ROM     3           53       159
103     Keyboard   1           17       17
---------------------------------------
The total of the invoice is : 176
```

What's the displayed result?

Q5). Write a function that takes an item code as a parameter and returns the amount saved.

Note: Amount saved = Sale Price – Sold Price.

Lab 6: Trigger

Q1). Write a trigger that displays "A record has been deleted" after deleting a record from the InvoiceDetail table.

Q2). Write a trigger that displays the message "You do not have the permission to delete a record" instead of deleting a record from the Invoices table.

Q3). Write a trigger that not allowed settling an item whose sale price does not exceeds 50% of the purchase price.

Q4). Add an Active attribute to the Items table whose possible values are yes and no. Write a trigger that sets to zero the quantity in stock of disabled items.

Q5). Write a trigger that rejects to add an item to an order whose quantity is larger than the stock.

Q6). Write a trigger that allows updating the stock after insertion, deletion or modification of a command line.

Q7). We wish to record in a table the invoices whose the date of purchase is not in the current month (deleted invoices from Invoices table).
For this, create a new table OldInvoices containing invoice number, purchase date, customer number and name, and the total amount of the invoice.
Write a trigger that preserves all deleted invoices in this table.

Lab 7 : Views

Q1). Using SQL, Formulate the following queries.

a. List of clients who ordered the item identified by 100.
b. Name of clients who ordered at least one item having price greater than or equal to 100.
c. Name of clients who have not ordered any item.
d. Name of clients who have not ordered the item identified by 101.
e. Name of clients who ordered all items.

Q2). Create a view named CliCoItem that allows visualizing the characteristics of items ordered by each customer (attributes to select: Number and name of clients, ID, Name and sale price of article).

Q3). List the contents of the CliCoItem view.

Q4). Reformulate the first two queries of question 1 using the CliCoItem view. Comment?

Q5). Formulate the following queries using the CliCoItem view.

a. For each customer, the price of the most expensive item that was ordered.
b. For each customer whose name ends with the letter 'e', average price of items ordered.
c. Maximum of Totals prices for all items ordered by different customers.
d. Numbers of products ordered more than twice.

Q6). Create a view called DiscountItemView allows to visualize the ID, Item and Discount price of discount item where the description is Discount.

Q7). Insert in DiscountItemView view the record (101, 50).

Q8). If you list the contents of the DiscountItemView view, does the tuple inserted in Q8 appear? Justify your answer

Q9). Write a trigger that fires after inserting a row in the DiscountItemView view and automatically adds a row in DiscountItem table, the value of the attribute Description is Discount instead (INSTEAD of) insert the row in the view.

Q10). Create a view named CliItem based on CliCoItem and allows displaying only the attributes Name of client and name of item. List the contents of the CliItem view.

Q11). Destroy the CliCoItem view. List the contents of the CliItem view. Conclusion? Destroy the CliItem view.

Lab 8: Cursors

Q1). Write a T-SQL script that determines all invoices (number and total) of the customer identified by 333.

Example 1 : Client 330	`This customer was not recorded in the database.`
Example 2: Client 765	`This customer has no invoice.`

Example 3 : Client 455	`The orders of customer 455 are:`
	`1 176`
	`4 51`
	`The number of orders is 2`

Q2). By using cursor, write a procedure that takes an invoice number as a parameter and displays the detail of the invoice as follows:

```
Invoice number: 1
Date: 1/1/2016
Code   Name      Quantity   Price   Total
100    CD-ROM    3          53      159
103    Keyboard  1          17      17
---------------------------------------
The total of the invoice is : 176
```

Q3). Display the Jack Brown customer invoices as follows:

```
Client: Jack Brown (455)

Invoice number: 1
Date: 1/1/2016
Code   Name      Quantity   Price   Total
100    CD-ROM    3          53      159
103    Keyboard  1          17      17
---------------------------------------
The total of the invoice is : 176

Invoice number: 4
Date: 1/1/2016
Code   Name      Quantity   Price   Total
103    Keyboard  3          17      51
---------------------------------------
The total of the invoice is : 51

The number of invoices is: 2
```

Q4). By using a cursor, write a T-SQL script that allows increasing all selling prices of items as follows:

- 4% for the first quarter of the highest prices.
- 6% for the second quarter of the highest prices.
- 8% for the rest of the items.

The script must display the name of the item and its price before and after the increase.